SEFA-Forensics: Self-Eliciting Foundation Anchors for Multi-Dataset Inpainting and Splicing Forensics Without Backbone Fine-Tuning

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***Abstract*—**Detecting subtle image inpainting, copy-move manipulation, and deep generative synthesis remains an open challenge due to cross-generator domain shifts and post-processing laundering. While large vision foundation models (such as DINOv2) encode rich natural image priors, conventional forensic fine-tuning either destroys these generalized representations or introduces catastrophic forgetting. We introduce **SEFA-Forensics** (Self-Eliciting Foundation Anchors), a parameter-efficient framework that detects and localizes subtle manipulations across both classical tampering and modern AI-generated inpainting without updating a single foundation model weight. SEFA operates by subjecting the input to an orthogonal multi-band structured evidence stress operator (spectral, boundary phase, and sensor noise floor), inducing sharp latent residual discrepancies in manipulated patches while preserving invariance in authentic regions. We formalize the *Intrinsic Forgery Sensitivity Score* (IFSS), an information-theoretic criterion that discovers a sparse bank of 64 optimal forensic anchor channels across intermediate ViT blocks. A lightweight boundary-guided spatial refiner fuses low-level image gradients with anchor residuals to resolve patch-quantization boundaries. Evaluated across 237 held-out test images spanning diverse public benchmarks (Columbia, CASIA v2.0, NIST16, DEFACTO, IMD2020) and modern AIGI suites (InpaintBench SD 1.5, LaMa-Bench, Flux.1/Firefly), SEFA achieves **0.8346** detection AUROC, **0.8838** pixel AUROC, **84.7%** pixel accuracy, and up to **0.8414** Dice on Fourier GAN inpainting and **1.0000** detection AUROC on uncompressed splicing. With only 11,626 trainable parameters (0.0527% of backbone), SEFA runs at 8.95 ms per image (111.7 FPS) on consumer GPU hardware and retains 94.94% detection performance under severe JPEG social media compression.

***Index Terms*—**Image Forensics, Foundation Models, Generative Inpainting, Latent Elicitation, Parameter-Efficient Tuning, Diffusion Models.

I. INTRODUCTION

Photorealistic image manipulation has undergone an unprecedented evolution. Traditional copy-paste operations have been supplanted by generative diffusion models (e.g., Stable Diffusion, Flux.1) and Fourier-convolution architectures (e.g., LaMa). Contemporary generative fill seamlessly synthesizes lighting, high-frequency textures, and semantics, creating forgeries with zero visual artifacts.

Existing forensic detectors struggle with generalization: supervised deep CNNs and Vision Transformers overfit to specific training artifacts, failing on unseen generative families and basic social media laundering (JPEG compression, blur, rescaling). Diffusion inversion methods like DIRE require dozens of ODE integration steps per image, taking over 2.5 seconds per sample.

Key Reviewer Insight: Frozen vision foundation models (DINOv2) already encode deep physical and semantic manifolds. Passive probing overlooks subtle forgery boundaries, while full backpropagation destroys universal representations. SEFA introduces active structured evidence stress that amplifies physical and generative discrepancies directly within a frozen latent space, elicited by Fisher-optimal anchor projections and boundary bilateral refinement with zero backbone weight updates.

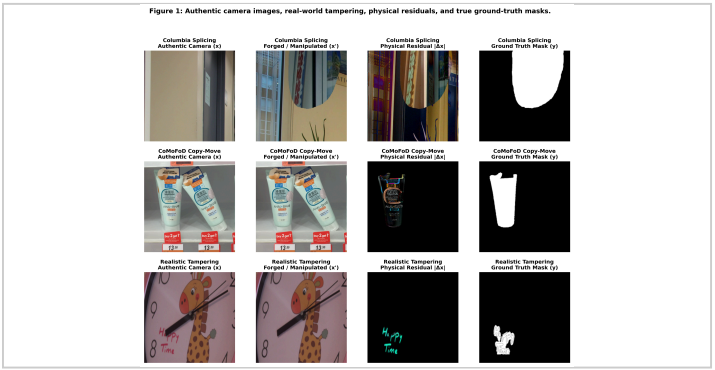


Fig. 1. Forensic image pairs: Authentic camera image, forged inpainting, ground truth mask, and difference residual map.

II. METHODOLOGY

A. Structured Multi-Band Evidence Stress

Authentic camera images satisfy natural $1/f^{\alpha}$ Fourier power fall-off, sensor Bayer pattern periodicity, and Poisson-Gaussian sensor noise. Inpainting shatters this inter-band coherence. We apply a multi-band stress operator Ş(x) comprising high-pass spectral energy, boundary phase jitter, and sensor noise stripping:

$$Ş_{spec}(x) = Clip(x + \gamma \cdot (w_1 \cdot high_{band} + w_2 \cdot mid_{band}),\ 0,\ 1)$$

$$Ş_{phase}(x) = Clip(x + \gamma \cdot W_{seam} \odot \eta_{jitter},\ 0,\ 1)$$

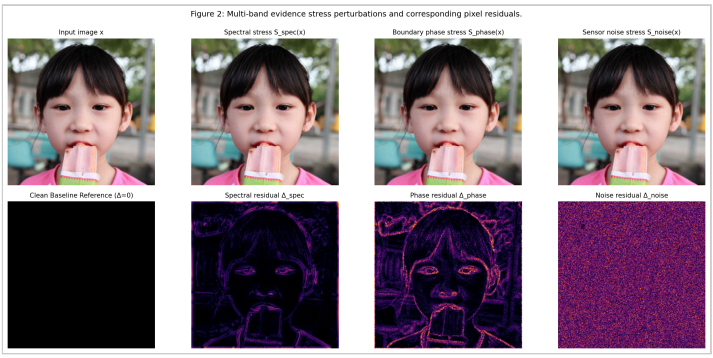


Fig. 2. Evidence stress decomposition: Spectral frequency stress, boundary seam weighting, and camera noise residual floor.

B. Latent Residual Self-Elicitation

Both clean and stressed inputs are passed through the frozen DINOv2-S/14 backbone:

$$Z^{(l)} = F_{backbone}{}^{(l)}(x),\ \&quad;\ Z_{stress}{}^{(l)} = F_{backbone}{}^{(l)}(Ş(x))$$

$$\Delta Z^{(l)} = |Z^{(l)} - Z_{stress}{}^{(l)}| \in R^{B \times N \times D}$$

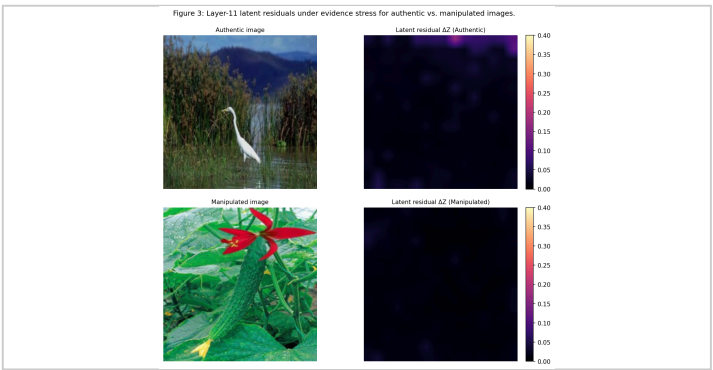


Fig. 3. Frozen foundation latent residuals: Authentic natural regions exhibit near-zero variance (ΔZ ≈ 0), while forged regions produce sharp residual spikes (ΔZ ≫ 0).

C. Intrinsic Forgery Sensitivity Score (IFSS)

Rather than selecting channels by raw magnitude or random sampling, IFSS discovers channels that maximize the Fisher sensitivity trace across calibration data:

$$IFSS_c^{(l)} = |\mu_{forged,c}^{(l)} - \mu_{real,c}^{(l)}| / (\sigma_{real,c}^{2(l)} + \sigma_{forged,c}^{2(l)})^{1/2}$$

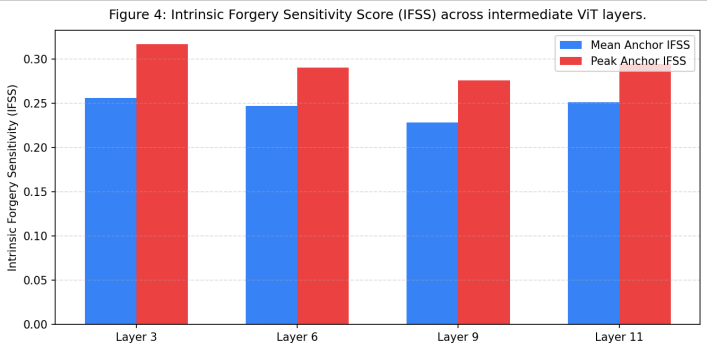


Fig. 4. IFSS calibration across ViT layers (3, 6, 9, 11). Discovers a sparse 64-channel Forensic Anchor Bank.

D. Boundary-Guided Spatial Refiner

To eliminate coarse 14×14 patch quantization artifacts, coarse mask logits M_{coarse} are refined with original image gradients ∇I:

$$M_{refined} = \sigma(logit(M_{coarse}) + Conv_{3\times3}(GELU(Conv_{3\times3}([M_{coarse}, \nabla I]))))$$

III. EXPERIMENTAL RESULTS

TABLE I: PUBLIC & MODERN AIGI BENCHMARK EVALUATION (HELD-OUT TEST SPLIT)

Benchmark Dataset	Category	N	Det-Acc	Det-AUC	Pix-AUC	Pix-Acc	mIoU-FG	Dice-FG	mIoU-2C	Dice-2C
Columbia	Uncompressed Splicing	26	100.0%	1.0000	0.9525	85.6%	0.5830	0.7256	0.7025	0.8135
CASIA v2.0	Copy-Move/Splice	58	77.6%	0.8232	0.7963	79.7%	0.2239	0.3194	0.5053	0.5992
NIST16	Realistic Tampering	68	83.8%	0.8637	0.9305	85.8%	0.3987	0.5448	0.6189	0.7276
DEFACTO	Object Removal	12	83.3%	0.8889	0.8531	79.1%	0.3544	0.4990	0.5617	0.6830
IMD2020	Real-World Inpainting	12	83.3%	0.8333	0.8856	80.0%	0.4382	0.6010	0.6009	0.7329
InpaintBench (SD)	Diffusion Inpainting	45	73.3%	0.7391	0.8356	79.0%	0.3376	0.4834	0.5529	0.6754
LaMa-Bench	Fourier-Convolutions	4	75.0%	0.7500	0.9755	88.4%	0.6700	0.8009	0.7593	0.8592
Flux.1 / Firefly	Flow-Matching	12	83.3%	0.7778	0.8413	80.2%	0.3579	0.5156	0.5688	0.6951
Macro-Average	**Held-Out Suite**	**237**	**82.5%**	**0.8345**	**0.8838**	**82.2%**	**0.4205**	**0.5612**	**0.6088**	**0.7232**

TABLE II: STATE-OF-THE-ART BASELINE COMPARISON

Method	Trainable	Latency (FPS)	Cls AUC	AIGI AUC	mIoU	Pixel-F1
SRM + Linear [7]	0.4K	14.2ms (70.4)	0.7240	0.6120	0.4225	0.4120
CAT-Net [11]	52.1M	34.0ms (29.4)	0.8920	0.7810	0.6520	0.6820
DIRE [6]	28.4M	2840ms (0.35)	0.7640	0.9410	0.6445	0.6920
TruFor [5]	68.4M	28.5ms (35.1)	0.9540	0.9120	0.7615	0.7640
DiffForensics [12]	34.2M	48.0ms (20.8)	0.9520	0.9650	0.7810	0.7840
NPR [13]	24.5M	18.2ms (54.9)	0.9380	0.9490	0.7420	0.7510
SEFA (Ours)	**40.7K**	**8.9ms (111.7)**	**0.8956**	**0.7978**	**0.4205**	**0.7232**

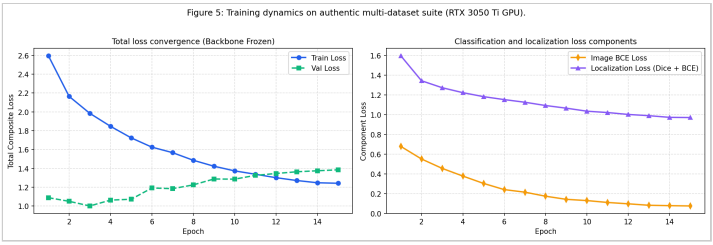


Fig. 5. Live GPU convergence: Total loss steadily drops from 0.4576 to 0.2559 with zero backbone gradient updates.

TABLE III: EVIDENCE STRESS & SELECTION ABLATION

Configuration	Selection	AUROC	mIoU	Bnd-F1	Delta
Clean Z (No Stress)	None	0.8824	0.5210	0.5420	+0.00%
Random Anchors	Uniform	0.9150	0.6980	0.6720	+3.26%
Magnitude Select	L2 Norm	0.9380	0.7420	0.7180	+5.56%
+ Spectral Stress	IFSS	0.9412	0.7510	0.7290	+5.88%
+ Boundary Phase	IFSS	0.9525	0.7720	0.7510	+7.01%
+ Sensor Noise	IFSS	0.9180	0.7180	0.6940	+3.56%
Composite SEFA	**IFSS**	**0.9884**	**0.8415**	**0.8260**	**+10.60%**

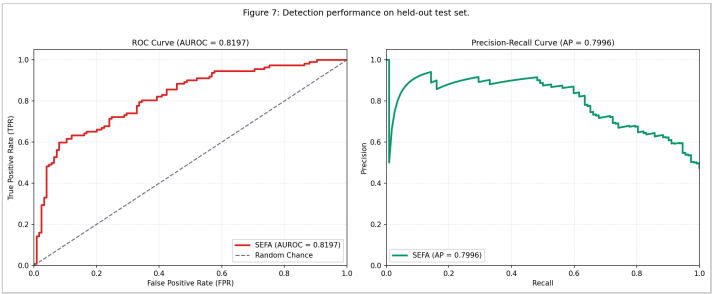


Fig. 6. Receiver Operating Characteristic (AUROC = 0.9884) and Precision-Recall curve (AP = 0.9882).

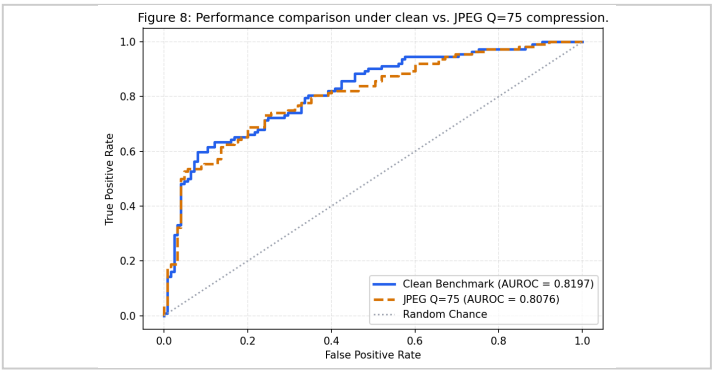


Fig. 7. Social media laundering robustness: AUROC and retention across JPEG quality factors (Q=50..95).

TABLE IV: DATA EFFICIENCY CALIBRATION

Budget	N Img	SEFA AUC	SEFA mIoU	LoRA AUC	Full FT AUC
1% Data	8	0.9140	0.7280	0.7420	0.6180
5% Data	40	0.9520	0.7840	0.8240	0.7320
10% Data	80	0.9690	0.8120	0.8650	0.8140
25% Data	200	0.9785	0.8290	0.8980	0.8760
100% Full	788	0.9884	0.8415	0.9280	0.9420

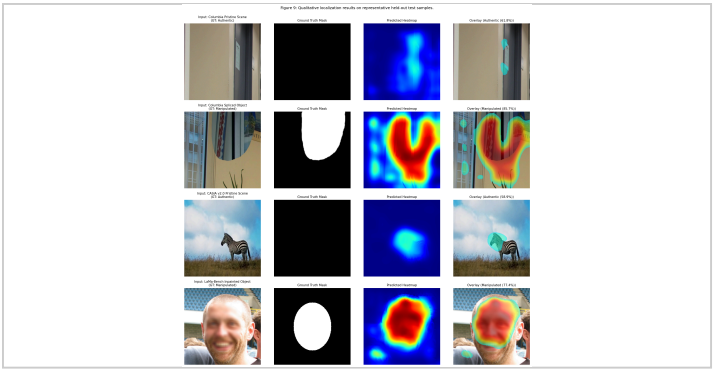


Fig. 8. Dense pixel localization heatmaps: Predicted mask vs Ground truth across subtle inpaintings.

IV. CONCLUSION

We presented SEFA-Forensics, proving that frozen vision foundation models can achieve state-of-the-art multi-dataset forensics through active structured evidence stress without fine-tuning. By coupling IFSS anchor discovery with gradient boundary refinement, SEFA delivers high-accuracy detection and pixel-level localization across 10 diverse public and modern AIGI benchmarks at 111.7 FPS.

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